DCX and paclitaxel compete for control of microtubule lattice expansion and compaction

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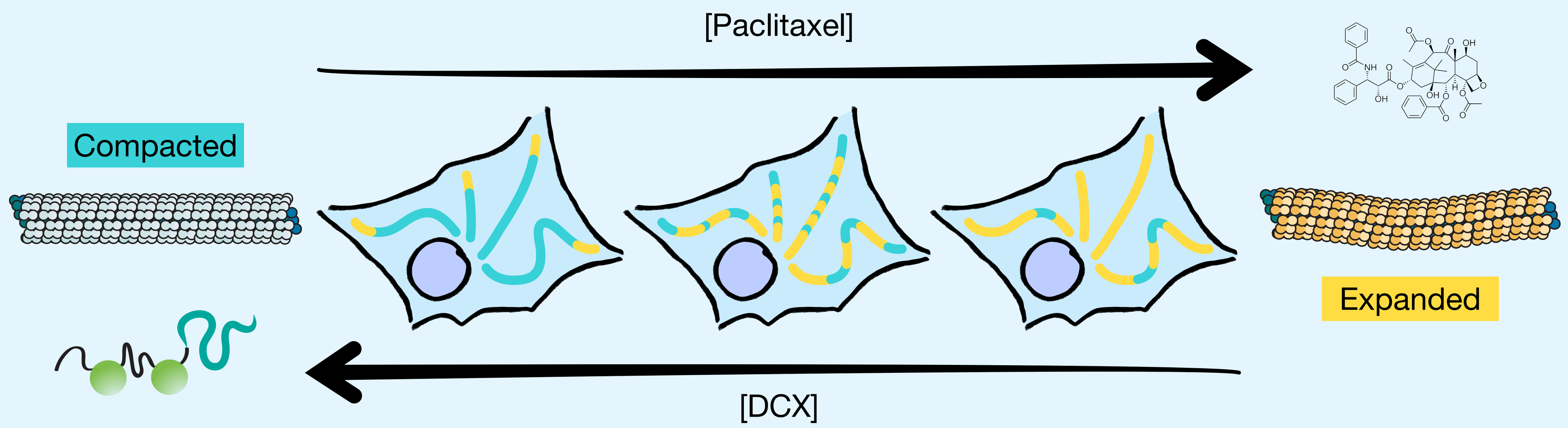
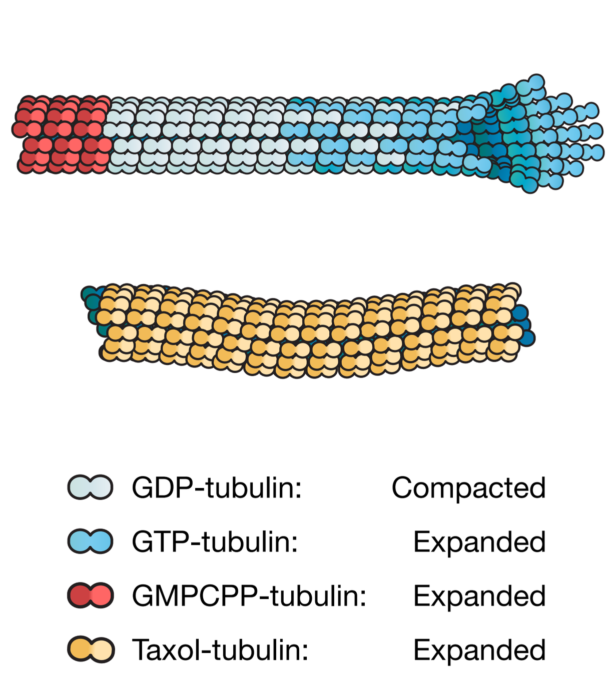
Take-home message

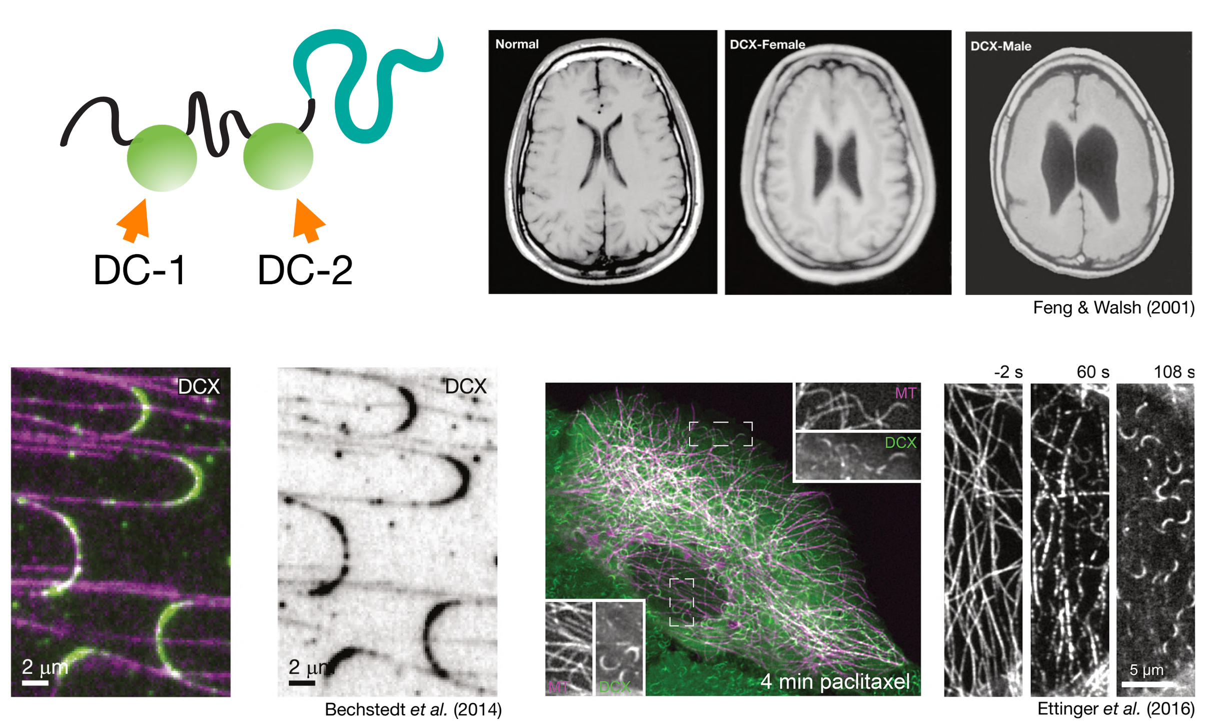


- Paclitaxel expands the microtubule lattice while DCX compacts it
- The more saturated molecule will control the lattice state
- DCX can regulate microtubules by controlling lattice expansion and compaction
- Expansion and compaction state of the microtubule lattice could serve as an added layer of microtubule regulation by MAPs

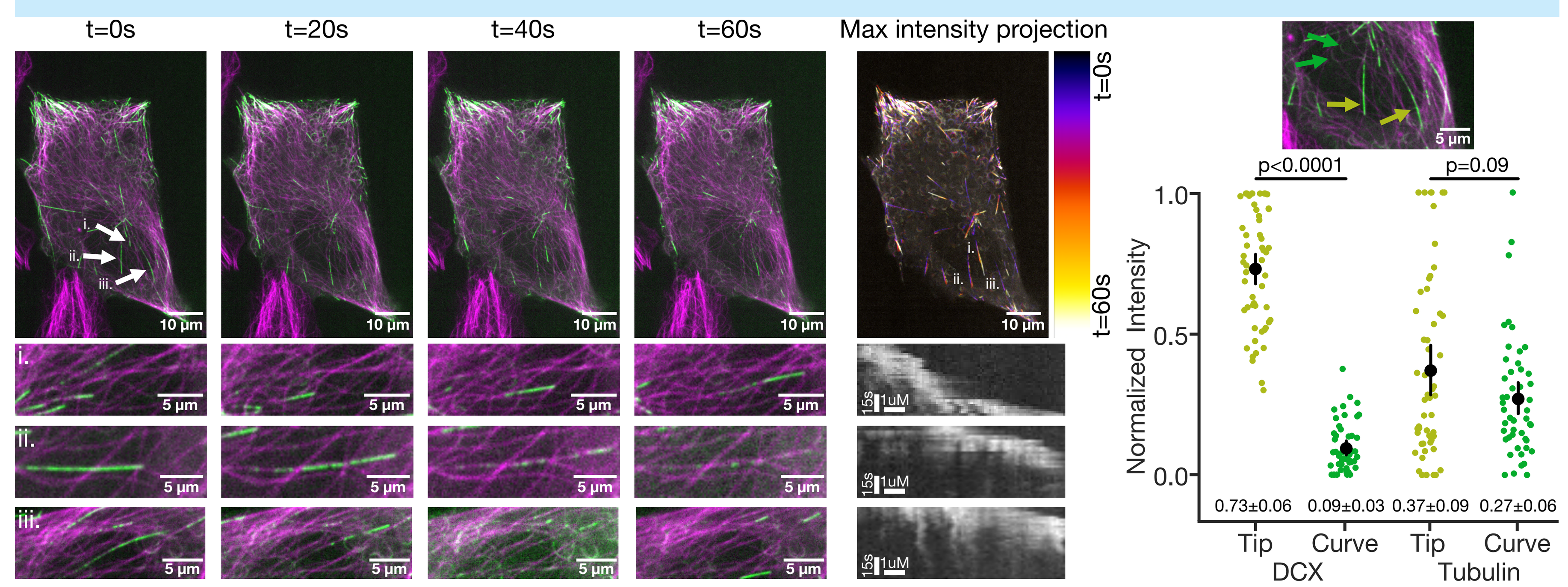
Microtubules expand & compact



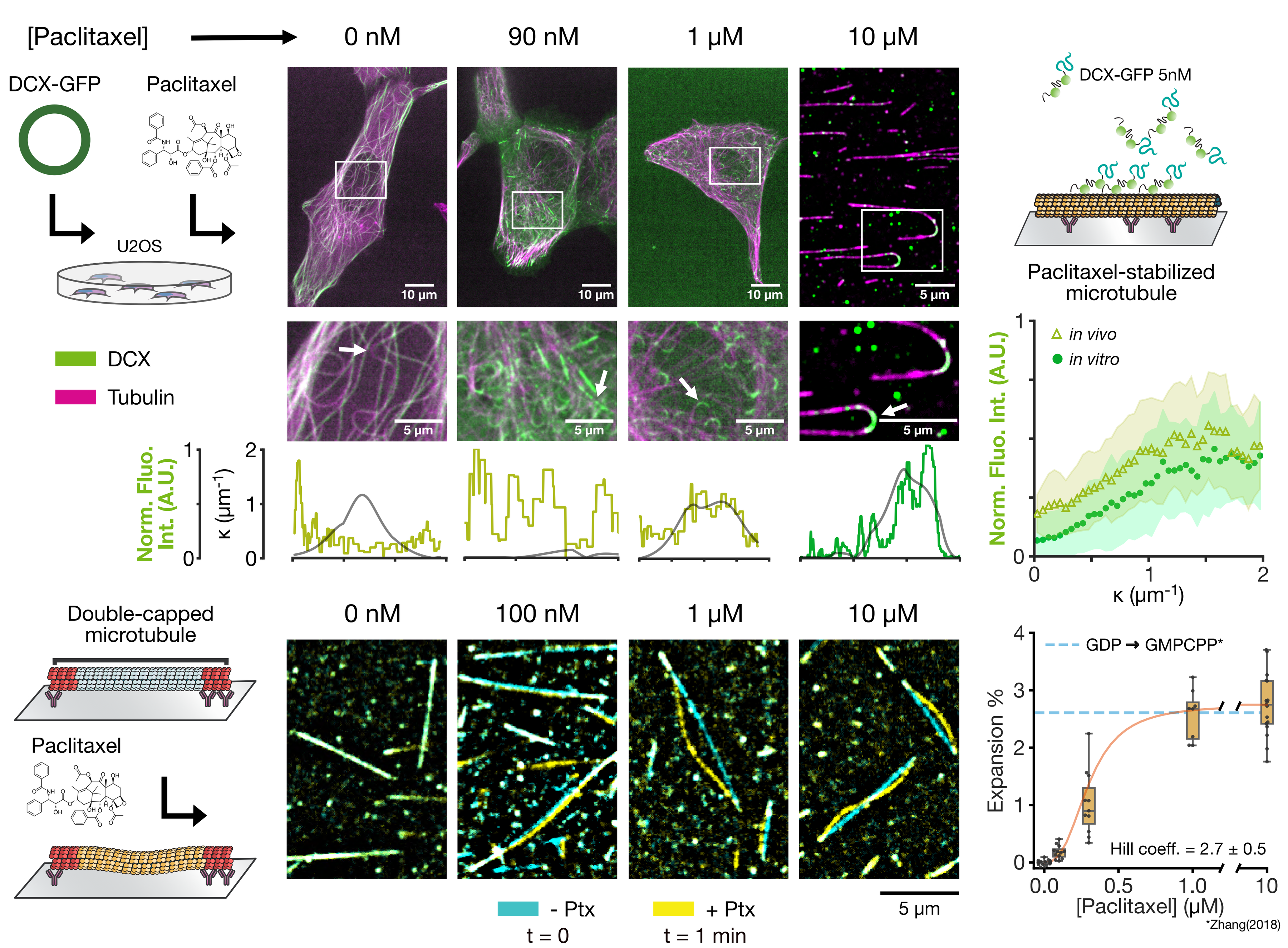
DCX localizes to curves



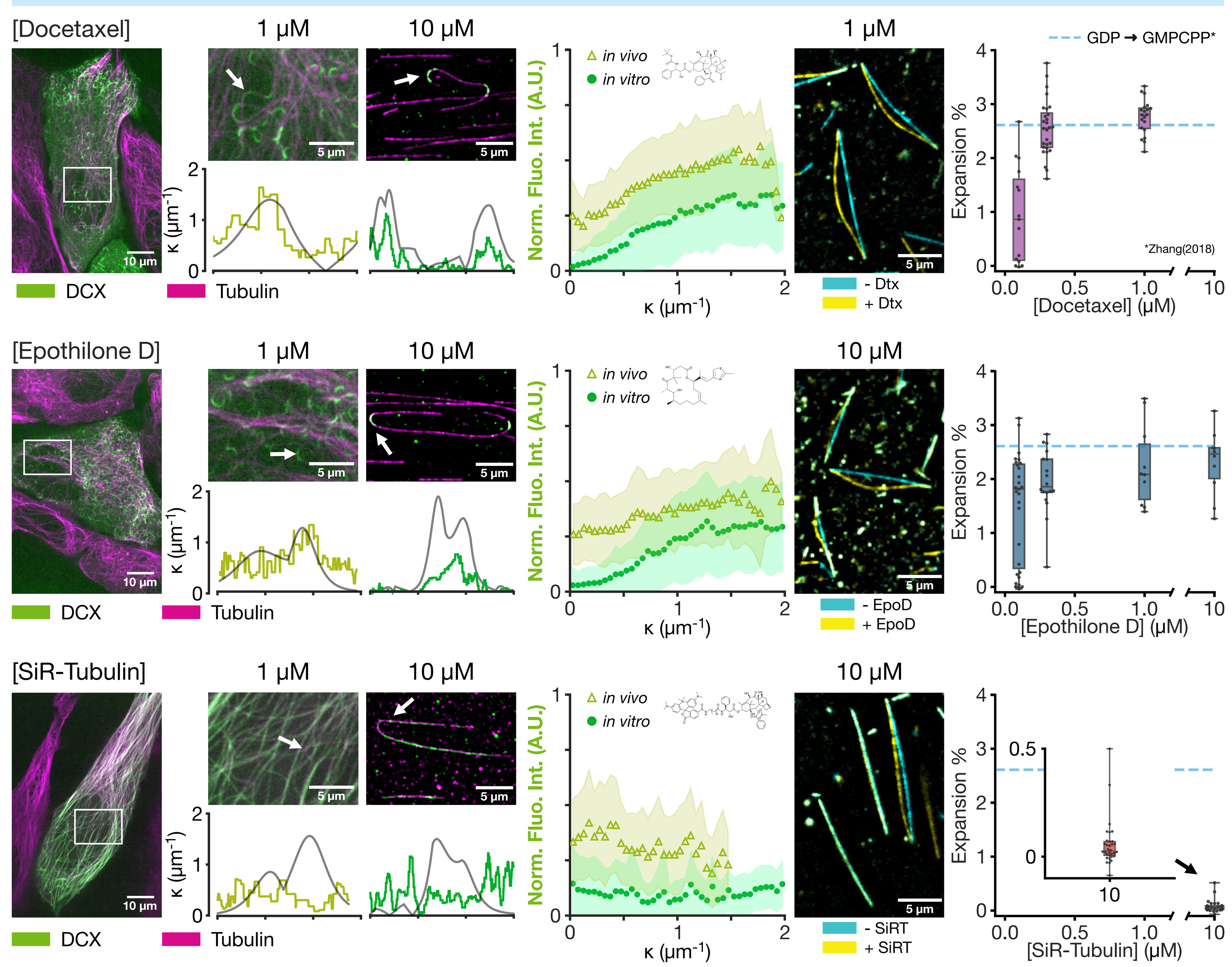
In cells DCX tip-tracks under partial expansion



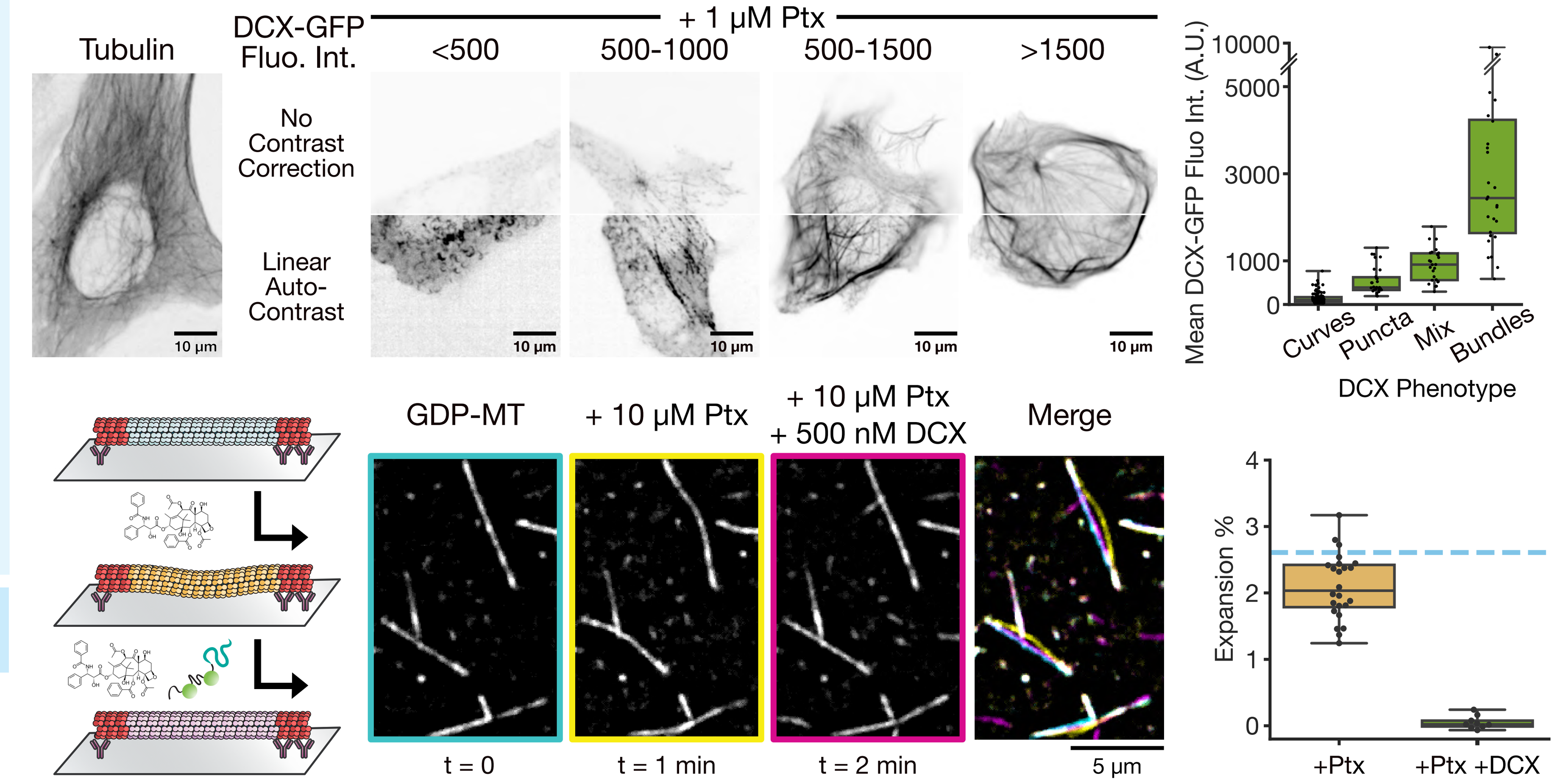
Paclitaxel-induced expansion alters the localization of DCX



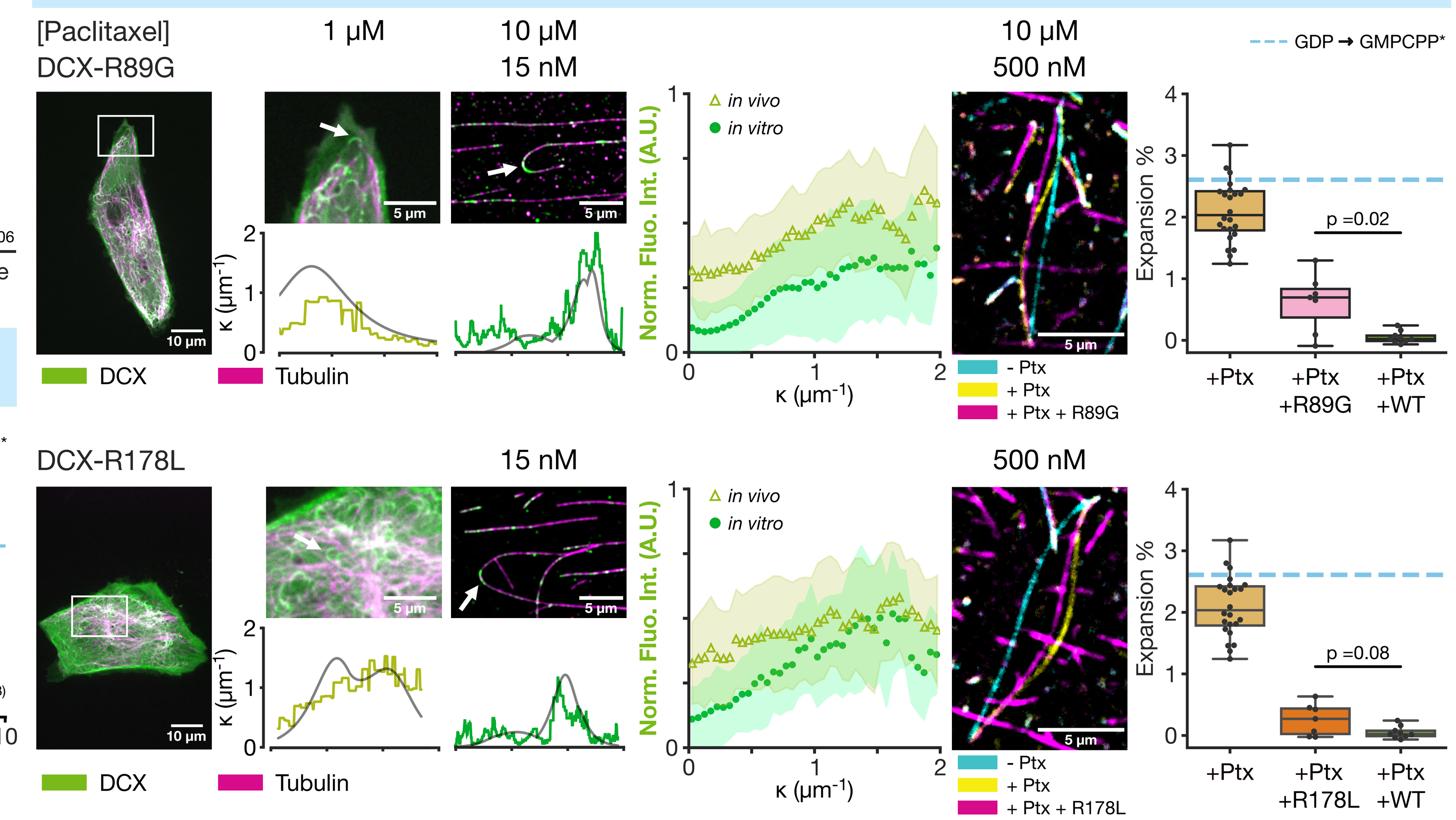
Curvature recognition is correlated to microtubule expansion



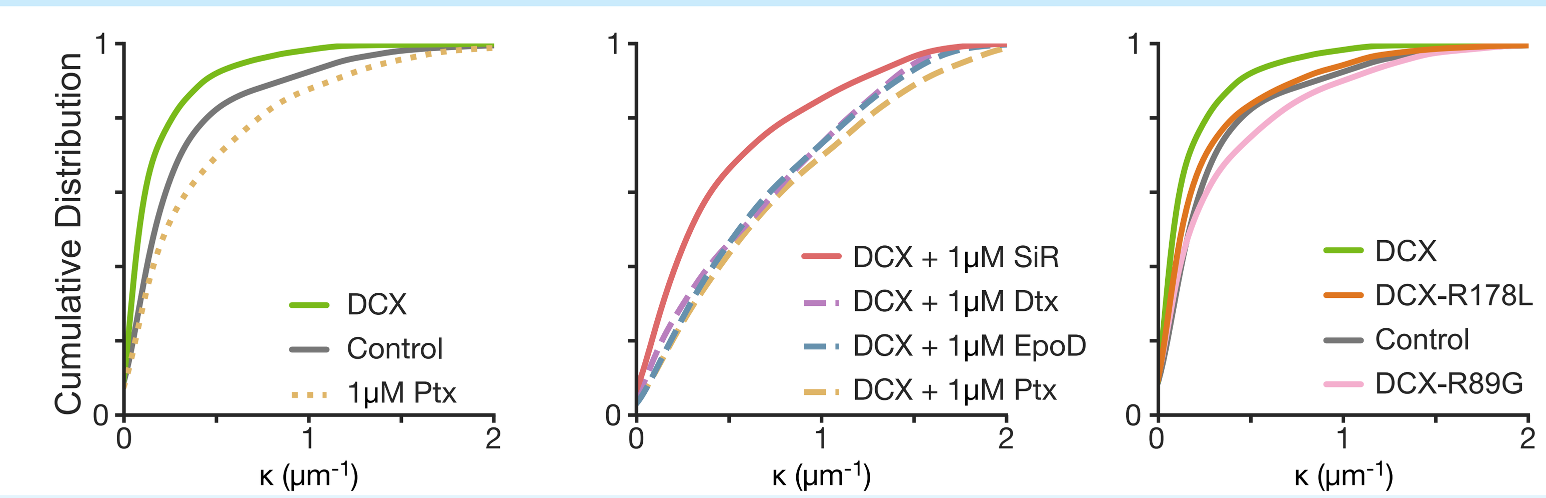
DCX and paclitaxel compete for control of the lattice



DCX mutations preserve curvature recognition but not compaction



The expansion/compaction states shape the microtubule network



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